



Society of Petroleum Engineers

SPE-230205-MS

Abandonment of an Inaccessible Well Using High-Risk Relief Well Perforation Techniques

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230205-MS](https://doi.org/10.2118/230205-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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ABSTRACT

Well abandonment operations present significant technical challenges, particularly when conventional intervention methods are unfeasible due to well integrity failures. In the case of Well-YX, operated by one of our customers in Oman, an integrity issue (Code 9) rendered the well inaccessible, with severe casing damage and an unidentified fish left in hole. The high risk associated with direct intervention led to the decision to drill a relief well targeting the lower section of the original wellbore through a complex sidetrack.

Initial plans involved milling through the 7" liner; however, the operation was unsuccessful. As an alternative, the team designed a novel approach to perforate the target well from outside the casing via the open hole section of the relief well. This methodology required a set of highly integrated solutions that addressed key challenges including:

- Ensuring penetration through both sides of the 7" casing,
- Creating sufficient entrance hole diameter to allow effective cement placement,
- Adapting to high dogleg severity and large borehole clearances,
- Utilizing only equipment available within the country.

Extensive coupon testing using 26-gram GH charges validated the required performance envelope. However, a 4.5" dummy gun run encountered a hold-up just 80 meters after running it inside the open hole. Multiple attempts to reach the desired depth failed, leading for further simulation modeling that demonstrated a successful push could be achieved using four heavy drill collar joints. A detailed phasing model was developed, highlighting the risk of missing the 7" casing entirely when using standard 60° phasing.

This operation required out-of-the-box thinking and the integration of mechanical, modeling, and operational expertise to overcome unforeseen downhole conditions. The solution succeeded in perforating the casing from the relief well, allowing for the secure isolation and abandonment of the original well.

In summary, this paper presents the innovative methodology, simulations, and execution strategy that helped achieve 20 shots on 7" casing in best case that enabled the successful abandonment of an inaccessible well using high-risk perforation techniques from a relief well, under extreme operational constraints.

BACKGROUND

Well-YX, located in the northern sector of the YX Shuaiba field, was originally drilled in 1991 as a deviated gas-lift producer targeting the Shuaiba formation. Over its life, the well underwent multiple workovers and completion changes in response to evolving production requirements and well integrity challenges. The first sidetrack (Hole-2) was drilled in 1998 to access additional reserves, followed shortly by a second sidetrack (Hole-3) in the same year. In 2013, the completion was converted from gas-lift to an electric submersible pump (ESP) configuration to improve production efficiency and lift performance.

In 2016, during a workover operation to replace a failed ESP, an obstruction was encountered at approximately 275 m inside the 7" scab casing. Milling operations were initiated to clear the blockage; however, progress stopped at 342 m. Cement recovered in the junk basket suggested a collapsed section of casing and poor annular isolation. To restore operational integrity, cement was pumped into the annulus to cover the obstruction, and a bridge plug was set at 200 m to isolate the damaged zone. A 3 tubing string was then installed, terminating at 180 m.

By late 2019, Well-YX was assigned Action Code-9 status following the identification of multiple well integrity failures. Notably, annular pressures particularly in Annulus D exceeded the Maximum Allowable Annular Surface Pressure (MAASP), indicating communication between annuli and possible barrier failures. These findings, combined with poor casing and cement conditions and the presence of an unidentified fish of unknown size and position, made the well inaccessible for standard intervention. In January 2021, a heavy hoist intervention was attempted to abandon the well, but the operation extended over 35 days without achieving the intended objectives, primarily due to severe downhole constraints, deteriorated tubular condition, and complex well geometry.

Given the accumulated integrity issues, limited accessibility, and the high operational risks associated with direct intervention, the decision was made to plan and drill a relief well (YX-RW). This well was designed to intersect the 7" casing of Well-YX at depth, enabling abandonment to be executed from outside the original wellbore while targeting a deep natural barrier for zonal isolation. The following sections outline the stepwise methodology, simulations, and field trials undertaken to design, validate, and execute the perforation program that ultimately achieved the required isolation, marking a first-of-its-kind operation in the history of perforation.

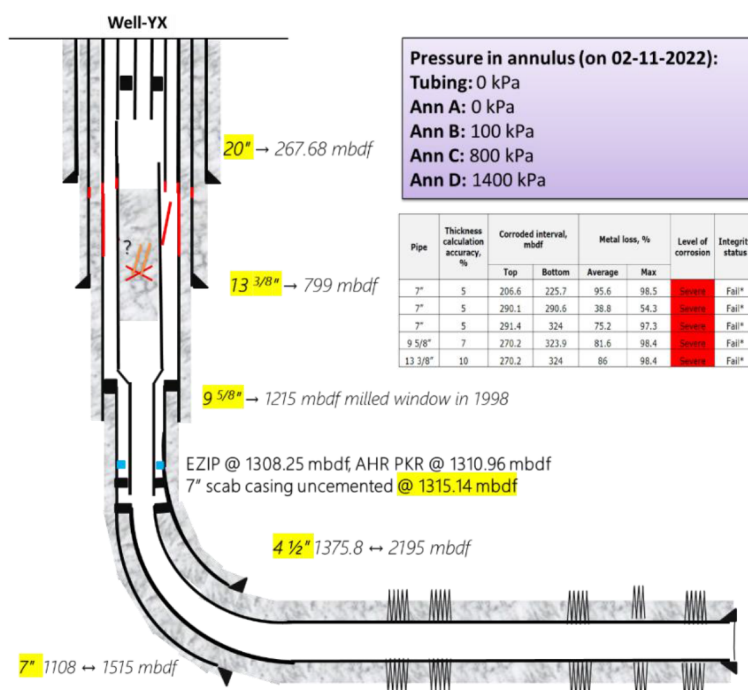


Figure 1—Well-YX schematic showing casing configuration

WELL OBJECTIVE

The primary objective of the relief well operation was to safely and effectively abandon Well-YX by establishing a permanent isolation barrier within the Nahr Umr formation. This would be achieved by perforating the 7" casing of the target well from the relief well trajectory, pumping cement into the perforated interval to create an effective seal, and ensuring long-term integrity in accordance with abandonment criteria.

A secondary objective was to design and implement a perforation strategy capable of penetrating both sides of the 7" casing and any associated cement sheath to enable adequate slurry placement. The methodology would need to account for challenging trajectory conditions, including high inclination and dogleg severity, limited open-hole clearance, and variable borehole stability.

An additional goal was to utilize only equipment and resources available within the country in order to minimize mobilization time and ensure operational continuity. This constraint required a detailed evaluation of all available perforating systems, shaped charges, and bottom-hole assembly (BHA) configurations to determine the most suitable combination for the well conditions.

Scope of Work

The planned operation involves drilling Hole-2 from the top hole of Well-YX-RW, following the defined trajectory and ranging strategy to locate, track, and intercept Well-YX at its 7" liner within the Nahr Umr formation. Upon interception, the 7" liner of Well-YX will be milled to gain internal access, enabling abandonment operations to be performed at the deep natural barrier using TCP perforation techniques. Following completion of these activities, both Well-YX and its relief well (Well-YX-RW) will be suspended and placed under observation for approximately one year, after which the planned shallow abandonment phase will be executed.

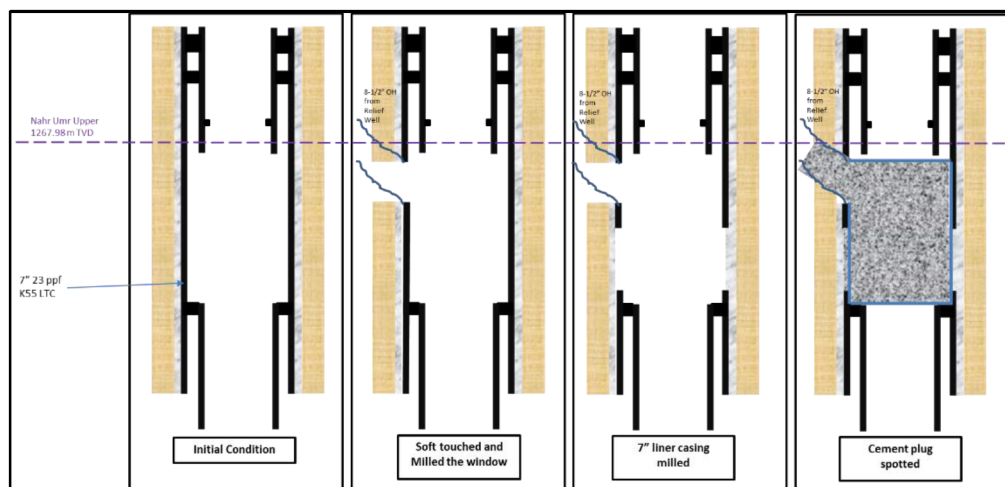


Figure 2—Summary of milling and cement placement sequence in the upper liner section

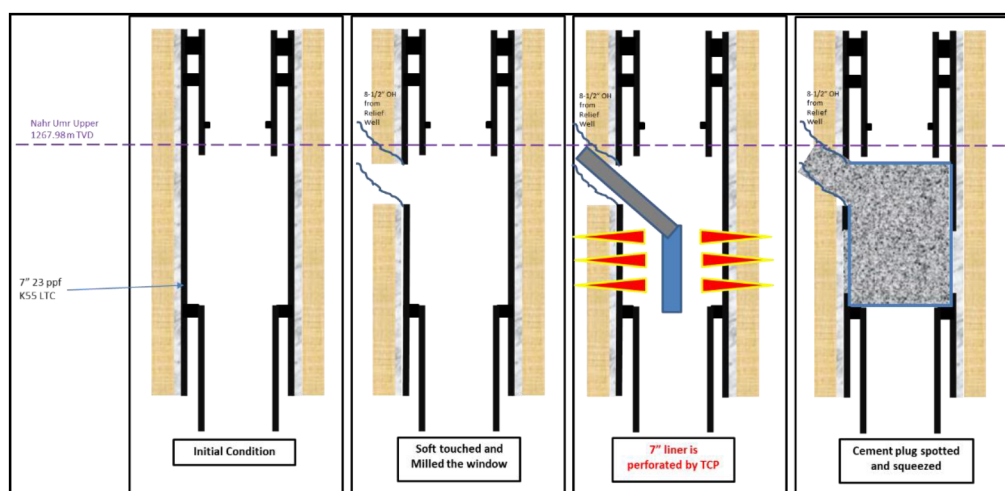


Figure 3—Summary of window milling, 7" liner perforation, and cement squeeze sequence.

Preparation

Given the well's compromised integrity and associated safety risks, extensive preparation and hazard mitigation measures were undertaken prior to commencing operations. A bleed-off line was installed and made ready as part of the pressure control contingency plan. The wellsite was fully barricaded to restrict unauthorized access and ensure compliance with safety protocols. Hydrogen sulfide (H_2S) detectors and pressure sensors were deployed around the cellar area and connected to all annuli, enabling continuous 24-hour monitoring of H_2S levels and annular pressure variations throughout the operation. These measures were implemented in recognition of the potential for hazardous gas release or uncontrolled pressure buildup during relief well interception activities.

Figure 4 shows the depth of the soft touch during drilling of the relief well. The green line represents Well-YX-RW, and the blue line represents Well-YX. The plan was to intercept and mill the well at 1335 m; however, due to the unreliability of the original deviation survey data from the target well, the soft touch occurred later than the expected depth. This made milling the 7-inch casing wall difficult because of the directional angle, resulting in continuous scratching of the casing wall while approaching the 4½" Top of Liner (TOL) depth at 1375.8 m. The chart also highlights the subsequent phase, which shows the total distance of unsuccessful interception attempts. Ultimately, interception of the target well could not be achieved.

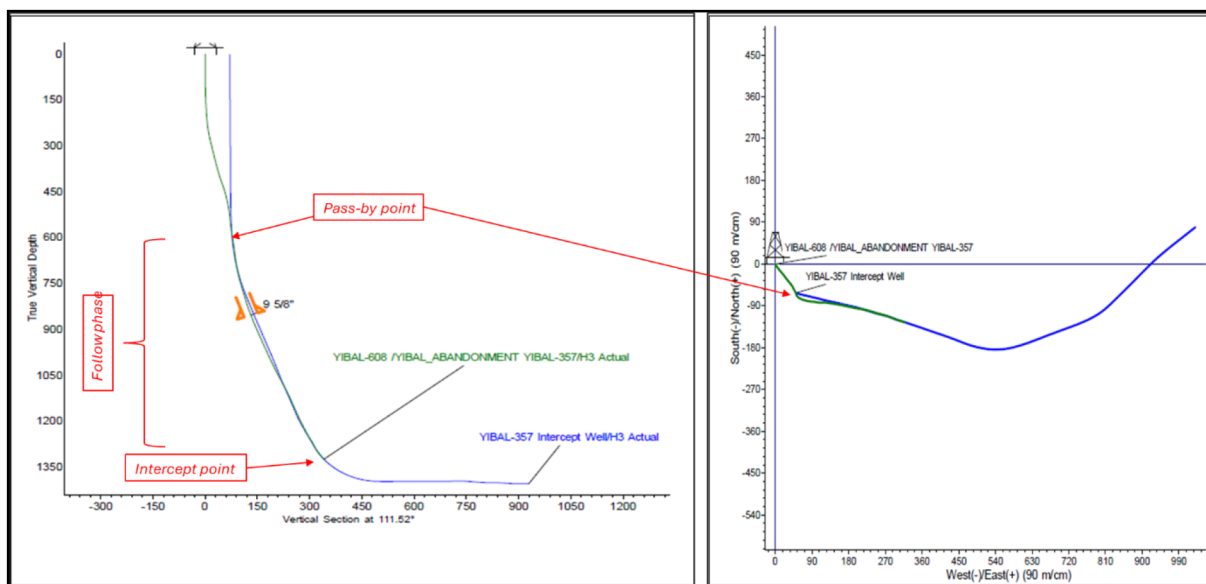


Figure 4—Well-YX and relief well trajectories with pass-by and intercept points.

Relief Well Design Considerations

Well completion setups in similar wells typically follow a 17 ½" hole section configuration common to the field's Shuaiba OP formation, beginning approximately 30 m into the Dammam formation and continuing to the top of the Natih formation. However, for this relief well, the design incorporated an S-shaped trajectory to meet the directional requirements of the planned interception, deviating from the conventional vertical profile.

Well-YX suffered a critical integrity failure first observed in 2019 through surface casing pressure differences. Diagnostics confirmed multiple structural deficiencies, including poor cement quality, degraded casing condition, and the presence of an unidentified fish in the wellbore at an unknown position. Multiple casing strings with varying levels of degradation further complicated the abandonment effort. These included: 20" casing set at 267 m, 13 ⅜" casing at 799 m, 9 ⅝" casing with a milled window at 1215 m, and an uncemented 7" scab casing at 1315 m. Annular pressure readings from November 2022 confirmed active pressure buildup, with Annulus D recording 1,400 kPa (203 psi).

Expected fluid losses in the Rus and UER formations were addressed using the standard losses decision tree applied in typical Shuaiba field operations. In the event that hole stability issues prevented forward progression, a contingency (fallback) plan was established to set the 13 ⅜" casing shoe within the Fiqi formation. The base case design specified an 8 ½" hole for the deep interception and a 12 ¼" hole for the shallow interception. In contrast, the fallback scenario incorporated a slimmer profile, 6 ⅝" for the deep intercept and 8 ½" for the shallow intercept to improve drillability under challenging hole conditions. Figure 5 below further illustrates both the 8 ½" base case and 6 1/8" fall back case scenarios.

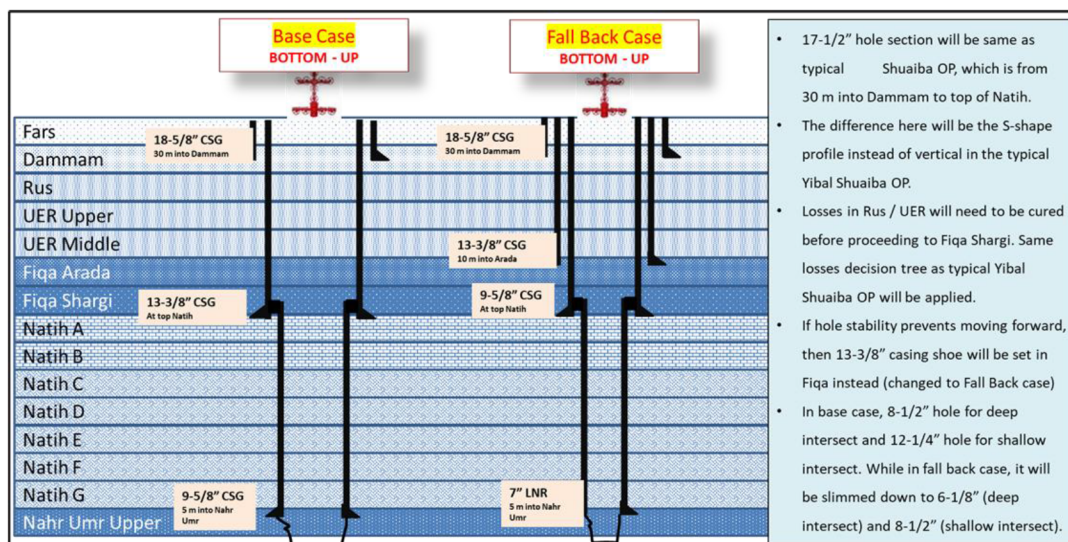


Figure 5—Base case and Fall-Back Case well designs with formation and casing scheme.

Perforation Design

Extensive evaluations and engineering reviews were undertaken to determine the most suitable perforation solution given the well's compromised condition. Investigations confirmed that direct re-entry into Well-YX was not feasible due to severely deteriorated casing integrity, the presence of an unidentified fish at an unknown position in the wellbore, and significant pressure buildup in Annuli B, C, and D, most notably 1,400 kPa recorded in Annulus D. These combined factors rendered any conventional re-entry intervention technically complex and high risk.

To mitigate these challenges, the decision was made to drill a relief well (YX-RW) and proceed through the open hole to intercept and mill the 7" casing wall of Well-YX, thereby creating access to the wellbore. The operational objective was to perform a perforation-based abandonment at a deep natural barrier using tubing-conveyed perforating (TCP) techniques. Following successful abandonment, both Well-YX and the relief well would be suspended and placed under a one-year monitoring program prior to the planned shallow abandonment phase.

The following section outlines the key operational challenges encountered in planning and preparing for this perforation program, providing the technical context for the stepwise design, simulation, and execution approach that followed.

OPERATIONAL CHALLENGES

The execution of the Well-YX abandonment via relief well perforation was defined by a unique set of operational, logistical, and technical challenges that required careful planning and adaptive execution. The constraints were not isolated to a single operational phase but were encountered from the initial planning stages through to the final perforation runs, with many factors interlinked and compounding one another.

Wellbore Accessibility and Geometry

The relief well's trajectory presented one of the most significant challenges during planning and execution phases. The first was wellbore accessibility: the relief well trajectory required precise ranging and interception under high dogleg severity up to 24° in some sections and inside a 6 1/8" open-hole environment. Such conditions increased the risk of tool hang-ups, restricted movement, and potential damage to perforating equipment during run-in-hole (RIH) operations. Figure 6 shows sketch of both Well-YX and the relief well with interception concept plan to access, perforate and pump cement across Nahr Umr formation to abandon the well.

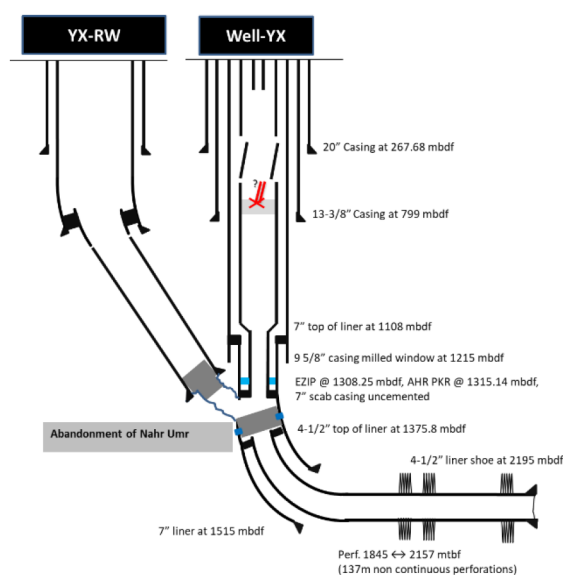


Figure 6—Relief well interception and abandonment of Nahr Umr section.

Survey accuracy presented another obstacle. Inconsistent gyro survey data between runs resulted in depth correlation uncertainty and made precise positioning of the milling relative to the target casing wall from Well-YX more complex. This inaccuracy increased the risk of missing the target or achieving proper access towards natural barrier target depth.

Target Casing Condition

The mechanical integrity of the target well's 7" scab casing was already compromised prior to the operation. Certain intervals were uncemented, others exhibited signs of collapse and metal loss were suspected in multiple sections based on historical integrity data. These conditions reduced the tolerance for perforation misalignment, as marginal shots may have been absorbed by deformed casing surfaces or misdirected into void spaces. Furthermore, the casing's lack of full cement bonding in some sections increased the risk of limited cement placement efficiency during the squeeze. Combined with the collapsed sections identified in earlier workovers, this significantly influenced perforating system selection and orientation strategies.

Equipment Availability and Lead Times

One of the key limitations was the unavailability of certain preferred perforating systems in-country. For instance, the 5" SLOT gun offering the largest area of flow (AOF) and the greatest potential for creating large cement entry points had a manufacturing lead time of approximately three months, which was incompatible with the operational timeline.

The technical advantage of the SLOT gun lies in its slot-shaped perforation holes with a 50% overlap in the phasing pattern. This design ensures full 360° coverage, unlike conventional guns that leave gaps between shots as seen in [Figure 7](#).

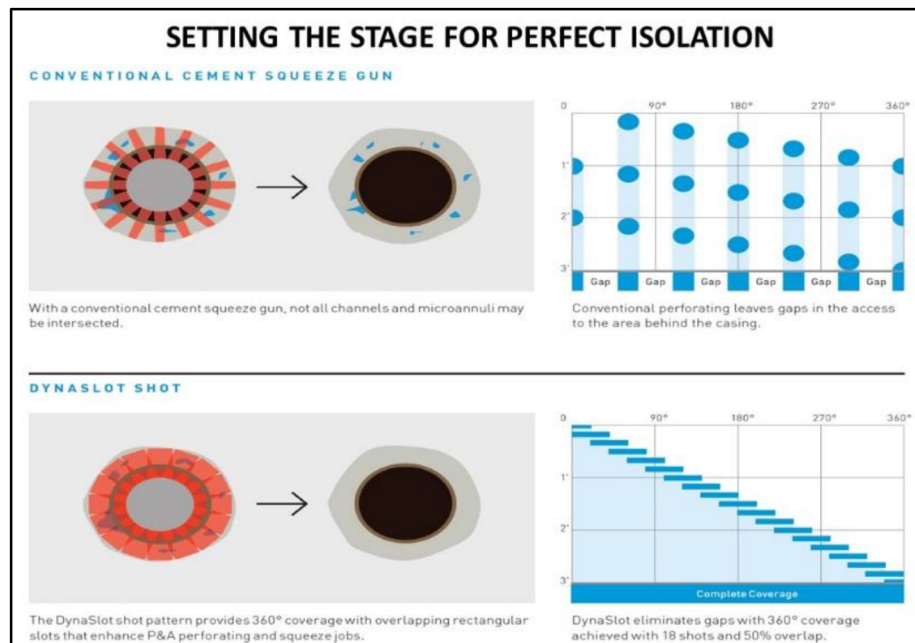


Figure 7—Conventional Cement Squeeze Gun Big Hole versus the SLOT gun. The Big Hole gun allows large area of flow, however the SLOT gun will provide 360° access to pump cement slurry.

The second-best option was the 4 ½" Big Hole gun with 12 SPF. While this configuration provides a reasonable open flow area to pump cement slurry, it does not deliver complete 360° coverage. Both gun options, however, are only viable once interception into Well-YX is confirmed. As illustrated in Figure 8, these two configurations were recommended as the best options to ensure proper abandonment. It is important to note that both the 5" SLOT and 4 ½" Big Hole guns are suitable only for single-casing applications and cannot be applied for side-wall perforation from open hole.

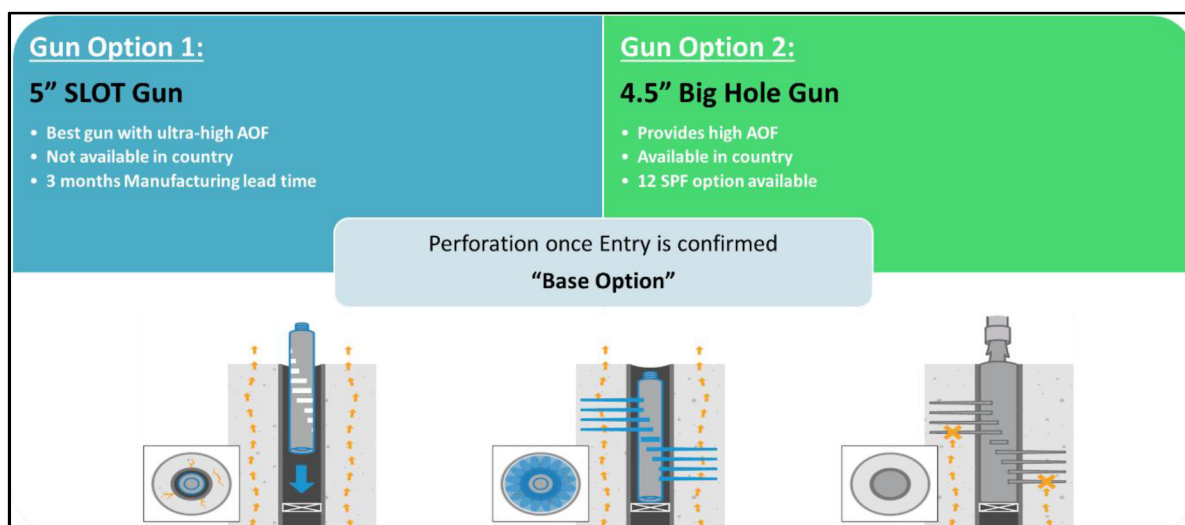


Figure 8—Comparison of perforation gun options for base case execution. Images at the bottom represent firstly the lowering of the 5" SLOT gun into the well, secondly perforating the well using 5 "SLOT gun with interval length covering the 360° access, and finally pumping of cement through the open slots and intersects all voids and channel, eliminating leaks and sealing the well.

During the planning phase, the 4 ½" Good Hole (GH) gun was not considered, as the interception success ratio was expected to be high. Consequently, no provisions were made for outer wall perforation at that stage.

However, due to concerns over gun availability, contingency options were developed using equipment already in-country in the event that interception was not achieved. This led to prioritization of configurations stocked in Oman, such as the 4 ½" GH gun with 6 SPF. This trade-off required additional engineering validation to confirm that the chosen configuration could achieve sufficient penetration and coverage under the prevailing well conditions.

Further details on the operational dynamics and final gun selection are presented in the methodology section.

String Configuration and Conveyance Limitation

Running large-diameter guns through the 6 1/8" open-hole section introduced specific mechanical challenges. Simulations revealed that slide forces were concentrated on the gun body rather than the drill pipe, indicating that extended BHA lengths could not be reliably pushed to target depth without risk of stalling. Dummy runs confirmed these findings, as the first 12 m gun assembly encountered a hold-up over 80 m short of the target depth. Addressing this required re-engineering of the string to incorporate heavyweight drill collars above the guns to improve weight-on-BHA transfer and minimize buckling in high-inclination sections.

Operational Risks During Execution

In addition to mechanical and logistical considerations, several high-impact operational risks were identified that could significantly affect the success of the abandonment program. These included the potential for severe fluid losses during cement placement, debris generation from perforating in degraded casing, and limited certainty regarding the final gun landing position prior to detonation.

The team needed to plan for both worst-case and best-case scenarios in terms of shot distribution against the target casing wall. In the worst case, misaligned phasing could result in zero contact with the casing, while in the best case, optimal orientation could deliver up to 20 shots on target over the designed interval. Bridging this performance gap required precise planning, robust simulation modeling, and the ability to rapidly adapt the perforation strategy based on intermediate results from dummy runs and coupon tests.

These combined challenges directly shaped the methodology that followed, necessitating iterative testing, simulation supported decision-making, and close integration of mechanical, modeling, and operational disciplines to achieve a successful abandonment outcome under extreme constraints.

METHODOLOGY

The execution of the Well-YX relief well perforation program followed a staged and adaptive workflow in which each operational phase directly informed the next. This approach was designed to mitigate the operational risks identified during planning, while accommodating the unique mechanical and logistical constraints of the project. Rather than relying solely on pre-job modeling, the methodology integrated in-country equipment validation, physical conveyance trials, coupon testing, and post-run simulation updates to progressively refine the perforation strategy.

The following subsections detail the stepwise process from initial dummy runs to final execution, highlighting the role of iterative testing, simulation-supported decision-making, and cross-disciplinary coordination in achieving the first documented side-wall perforation from an open-hole relief well in perforation history.

Adaptive Perforation Strategy Development Based on Field Challenges

During the running of the 7" liner under total losses, the assembly encountered a significant obstruction near the base of Natih-D at a shale interval. Multiple attempts to pass this point, including pulling out of hole (POOH) and re-running with a rotatable liner, were unsuccessful. As a result, the liner was retrieved

and the plan revised to drill a 6 1/8" section. This section presented its own set of operational challenges, particularly with trajectory reliability.

Gyro survey inconsistencies were a major concern, with significant variance in readings taken at the same stations across multiple runs. This led to uncertainty in the relief well trajectory and an increased risk of missing the intended interception with the target well. A soft contact with the target well's casing occurred at zero incident angle during drilling, which was attributed to inaccurate survey data from the target well. There was also a recognized risk of unintended interception leading to oil-based mud (OBM) losses.

Operational delays were also encountered during the rig move when a frac tank containing high-performance water-based mud (HPWBM) was found to have significant chemical settling due to prolonged stagnation. The tank's location beneath the rig's mud pumps prevented direct extraction by the vacuum tanker. This was resolved by deploying additional frac tanks for backup OBM and HPWBM storage and engaging the vendor's team to remove the settled material using a fire pump hose through the tank's side access door.

To improve survey reliability, multiple vendors were trialed, and station times were increased to enhance accuracy. For milling operations, a 3° motor bend and various milling assemblies, including Barracuda and concave mills, were used. However, due to limited accessibility, the operation shifted toward perforation as the preferred method to access the target well.

The original plan called for milling the 7" casing at 1,335 m, but this could not be achieved, as the bit was only scratching the sidewall without penetrating the casing. As the 4 1/2" top of liner (TOL) depth was approached, the decision was made to prepare and mobilize 2 7/8" deep-penetration guns. The aim was to perforate a sufficient interval to allow cement slurry placement, with part of the guns positioned inside the 4 1/2" liner and the remainder inside the 7" casing. Due to the change in gun type from big-hole to deep-penetration charges, additional runs were required to achieve the desired area of flow, leading to a double-shooting plan.

Since it was no longer possible to install the 7" liner, perforation had to be executed directly in the open hole. This introduced further risk due to high inclination and severe dogleg severity (24° DLS) within the 6 1/8" section. Running 4 1/2" big-hole guns in these conditions carried a high risk of sticking, prompting a more conservative choice of smaller 2 7/8" guns. This configuration was supported by further analysis and simulation, which confirmed the feasibility of running through the open hole and achieving contact with the target well's 7" casing.

However, the planned interception depth with the target casing was missed, making it impossible to execute the original plan of placing cement slurry from inside the well. As milling of the 7" casing was not successful, the final operational solution was to perforate the casing externally through the side wall from the open-hole section.

Intercept and Soft Wall Touch Confirmation

The intercept phase was complicated by several operational challenges. A stuck 13 3/8" casing hanger required fishing, cold cutting, and staged retrieval, while the use of a 21 1/4" BOP created logistical constraints due to heavy crane requirements. Severe losses were also encountered in the Rus formation, where unexpected H₂S and LEL releases occurred at shallow depth. Despite seven cement attempts, circulation could not be restored, and the decision was made to drill ahead under losses and later isolate the Shargi with 9 5/8" casing. This adjustment slimmed the well profile and ultimately required interception at 6 1/8" instead of 8 1/2".

Further complications arose from survey uncertainty. Early ranging runs showed the target well survey was of poor quality, with positional shifts exceeding 4 m compared to the original data. The drilling program had not initially highlighted the need to run Gyro with every ranging operation, which introduced early placement doubts. This was later corrected by updating the survey at each ranging run, with the WellSpot team confirming offsets and gradually improving accuracy. Ultimately, both ranging and Gyro tools were run

together in a dedicated non-magnetic ranging BHA, eliminating interference and ensuring reliable positional data.

With these issues resolved, the relief well advanced into the 6 1/8" open-hole section, where a soft touch was observed at 1378.3 m against the 7" casing of the target well. The ranging tool confirmed proximity at approximately 1 cm, while circulation brought metallic debris to surface consistent with 7" casing material. Together, these independent indicators provided reliable confirmation of wall touch, even though a full intersection was not achieved at this stage.

Perforation Gun Selection and Evaluation

In preparation for the planned interception, a 2 7/8" deep-penetration gun with 6 SPF was mobilized to the wellsite for perforation from inside the target well (Figure 9). After the interception attempt was unsuccessful, the option of repurposing the same gun for side-wall perforation in the 6 1/8" open hole was assessed. However, simulation results showed a low probability of consistently striking the 7" casing wall, and the gun was removed from consideration despite being available on location. Consequently, alternative gun sizes available in-country were evaluated, as the drilling unit required immediate mobilization of materials to maintain the operational schedule. The team was tasked with selecting the most suitable configuration from available resources to maximize the likelihood of success.

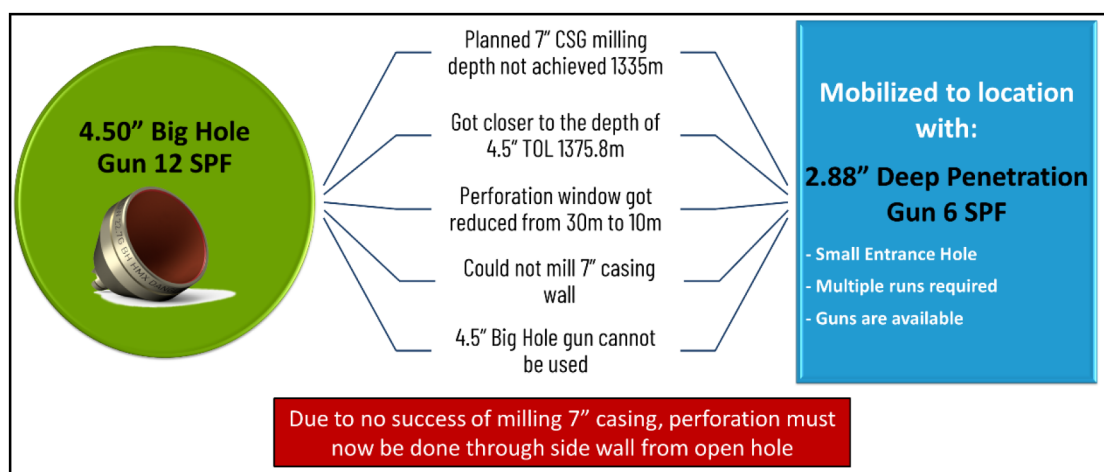


Figure 9—Development of perforation strategy from planned milling to failed interception of the 7" casing wall, to the final side-wall perforation concept in open hole.

Comparison of Available TCP Guns for Side-Wall Perforation

1. 2 7/8" 6 SPF, 15g Deep Penetration Charge
 - o Small Entrance Hole Diameter (EHD).
 - o Low chance to hit 7" casing walls.
2. 5.00" Oriented Gun 8 SPF, 26g Good Hole Charge
 - o Best Gun with good EHD and Phasing.
 - o Requires GYRO to apply orientation.
 - o High risk of getting stuck inside 6 1/8" open hole.
3. 4 1/2" 6 SPF, 26g Good Hole Charge
 - o Good EHD.

- Available in country.
- Requires coupon test.
- Requires dummy run due to high DLS.

Based on this evaluation, the 5" oriented gun offered the most favorable performance on paper but could not be implemented due to orientation and operational risks. The 4 ½" gun was ultimately selected as the most practical option, with additional testing (coupon shot and dummy run) planned to confirm suitability and ensure deployment to the required depth without restriction.

Good Hole Gun Performance Evaluation

Following the selection of the 4 ½" GH gun as the preferred perforating system, a detailed evaluation was performed to verify its suitability for the target well geometry and intended perforation objectives. The assessment considered tool clearance, shot spacing, and the positioning of perforation intervals to ensure optimal coverage while avoiding unintended damage to adjacent well components.

The perforation interval was strategically positioned to avoid both the 4 ½" top liner and the section below the top milling depth, ensuring no unintentional damage to existing barriers. A 10 m target zone was selected to provide sufficient coverage for effective cement placement and maximize the probability of achieving isolation. The clearance between the gun OD and the 6 1/8" hole, considering the maximum void, is 1.625" under the worst-case condition. The distance between two adjacent shots is 2.355". To prevent unintended perforation of the 4 ½" top liner at 1,390 m, as well as the section located below the top milling depth at 1,375 m

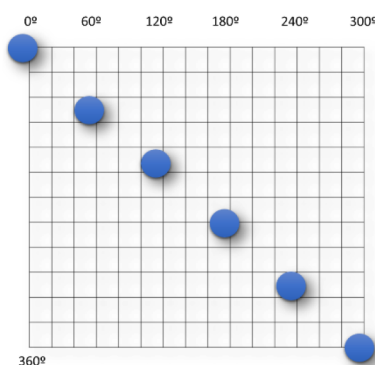


Figure 10—4 ½" 6 SPF Gun Phasing Shot Alignment pattern

To maximize the probability of contacting the 7" casing in Well-YX, a shifting-shots-between-guns approach was applied. Three joints of 4 ½" guns (3 m, 3 m, and 6 m) were deployed in a single run, each oriented with a 20° offset from the joint below. This arrangement reduced shot spacing and achieved a combined phasing of 20° over the 10 m interval, compared with the standard 60°, thereby increasing the likelihood of multiple impacts per foot on the 7" casing as shown in Figure 11.

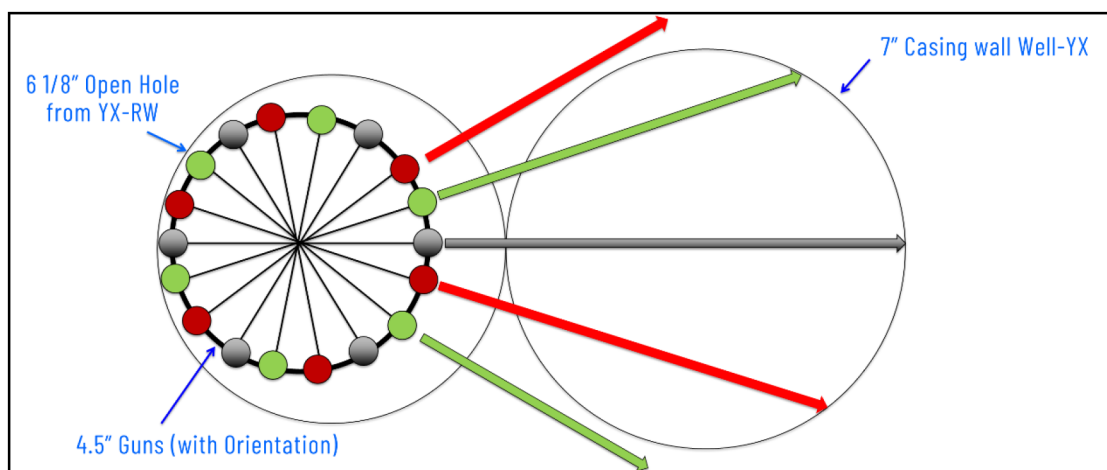


Figure 11—un orientation pattern illustrating increased probability of striking the 7" casing. Each color represents one of the three connected guns, ensuring at least one aligns with the target well.

Gun orientation was achieved using special oriented tandems (Figure 12), which were available in-country from a separate project. These tandems incorporate a swivel and locking mechanism that enables controlled angular offsets between guns. The applied configuration introduced a 20-30° shift from Gun 1 to Gun 2, followed by an additional 20-30° shift from Gun 2 to Gun 3.



Figure 12—Orienting Gun Tandem are used to apply 20° phasing offset concept between guns, after applying orientation the tandems are locked and will not rotate while RIH.

A 26g Good Hole shaped charge coupon test was conducted to simulate the actual downhole conditions, maintaining the same clearance between the gun OD in the 6 1/8" open hole and the 7" casing. Brine was placed inside the 7" casing, with cement positioned on the opposite side to represent the annular barrier. Two shots were fired to validate penetration performance, and both yielded positive results.

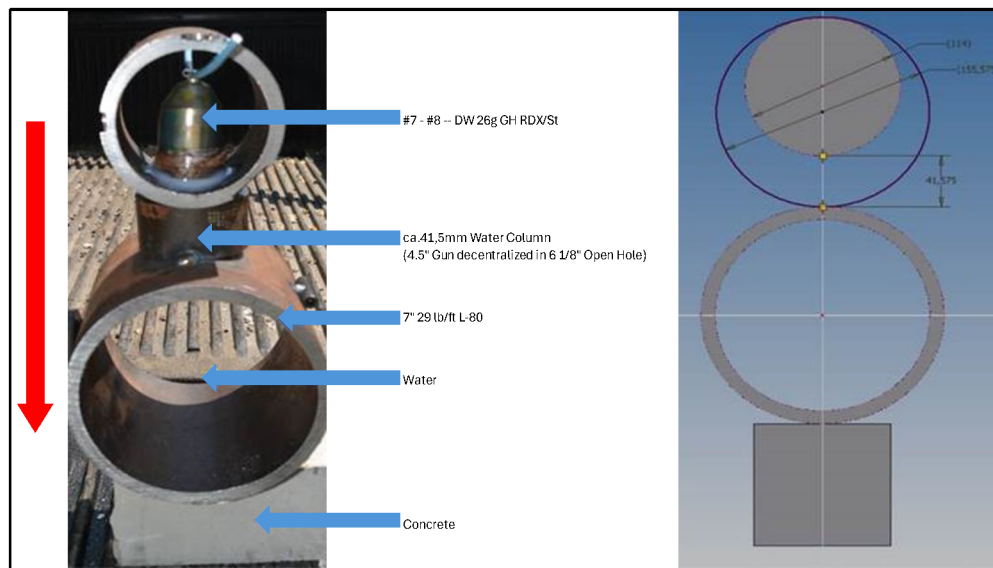


Figure 13—API 19B Section 2 coupon test setup for the 26g Good Hole shaped charge simulating downhole conditions. The gun was positioned at the far side of a 6 1/8" open hole and fired against a 7" casing wall backed by a 5" concrete block.

In both tests, the 7" casing (29 lb/ft) was successfully penetrated from opposite sides, along with more than 5" of cement behind the casing, limited with the cement block positioned during the test setup, See Figure 14 for EHD and penetration results. Based on these results, a 20° orientation between guns is recommended to maximize the number of shots impacting the 7" casing wall in the target well Y-X.

Test #	EHD – 7" (1) (inch)	EHD – 7" (2) (inch)	Penetration (inch)	Comments
7	0.35	0.193	> 4.9	Penetration in Concrete
8	0.366	0.193	> 4.9	Penetration in Concrete

Figure 14—Coupon test results show Entrance Hole Diameter in 7" casing walls. Penetration on cement behind casing was 125mm (5 inch) as this was the total thickness of the concrete used in the test.



Figure 15—Photos of the 7" casing showing successful penetration through both walls by the 26g GH shaped charge.

Perforation Strategy Development

The development of the perforation strategy for Well-YX was driven by the need to reconcile multiple constraints:

- limited access to the target well,
- degraded tubular conditions,
- high annular pressures,
- and uncertainty in final gun positioning.

Under such conditions, a single, fixed design carried an unacceptably high risk of failure. Instead, the team adopted a progressive, evidence-based approach in which each stage of planning and execution was informed by the verified performance of previous stages.

Rather than treating dummy runs, simulations, and coupon testing as isolated checks, these activities were integrated into a continuous feedback loop. Mechanical performance data from field trials influenced subsequent modeling assumptions, while simulation results guided modifications to gun length, phasing, and string configuration. This closed-loop workflow ensured that each decision whether related to tool conveyance, orientation strategy, or interval coverage was supported by both empirical evidence and predictive analysis.

The following five steps outline the sequence of trials, simulations, and design refinements that culminated in the first documented side-wall perforation from an open-hole relief well, establishing communication across two walls of the 7" casing and enabling successful abandonment under extreme operational constraints.

Step 1 - First Dummy Run: 12 m (40 ft) Gun Assembly

The dummy run assembly (Figure 16) encountered a hold-up at 1,306.78 m within the open hole, preventing access to the planned target depth of 1,388 m to tag the 6 1/8" hole total depth (TD). Multiple reciprocation attempts were made, but the same obstruction was encountered on each pass. Surface overpulls of 2-4 tons were applied without success. The run was therefore terminated, and the assembly was pulled out of hole (POOH) to evaluate alternative options.

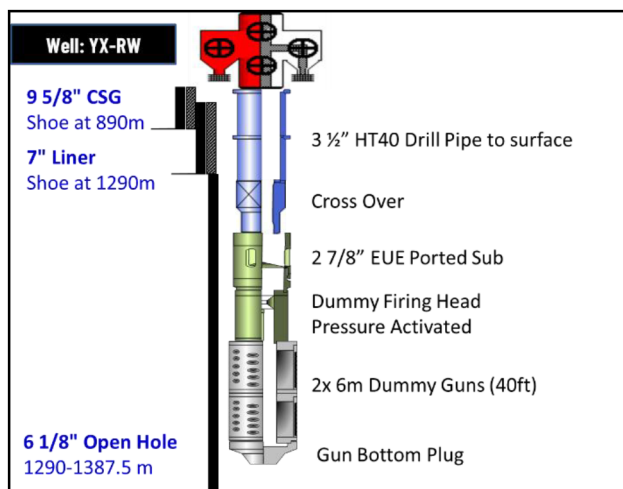


Figure 16—Bottom-hole assembly setup for TCP during initial dummy run in Well YX-RW.

Step 2 - EDM Simulation and String Re-Design

Post-run analysis was conducted using EDM simulation software to evaluate the conveyance performance of the 4 gun within the 6 1/8" open hole. The simulation (Figure 17) with a 12 m gun indicated that the highest slide forces were acting on the gun body rather than the drill pipe, limiting weight transfer to the string and increasing the risk of damaging the guns. Pushing the 4 1/2" guns into the 6 1/8" hole with a 3 1/2" drill pipe required significant force under these conditions.

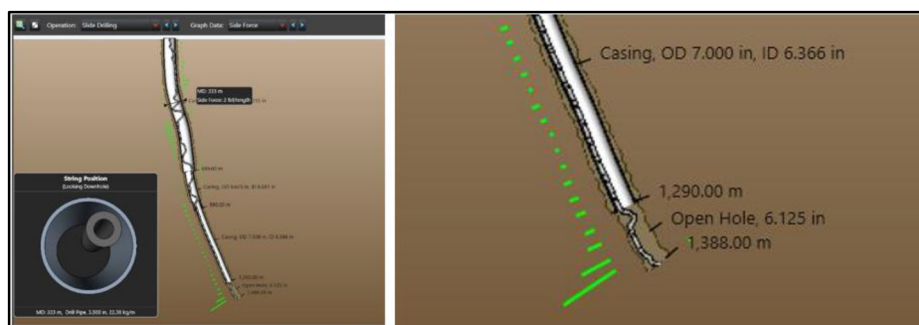


Figure 17—Simulation Software to examine RIH guns with different string configurations to identify drag forces.

Introducing a heavy-weight drill collar (HWDC) above the gun in the simulation (Figure 18) significantly improved sliding force performance. Various gun lengths 18 m, 12 m, and 6 m were modeled, and the results showed that four joints of HWDC provided the best capability for delivering the guns to total depth (TD). The simulation confirmed that, with this configuration, the slide forces acted primarily on the HWDC, enabling smoother force transfer and allowing the guns to be advanced to the intended depth. Based on these findings, four joints of HWDC were required for the planned live perforation runs.

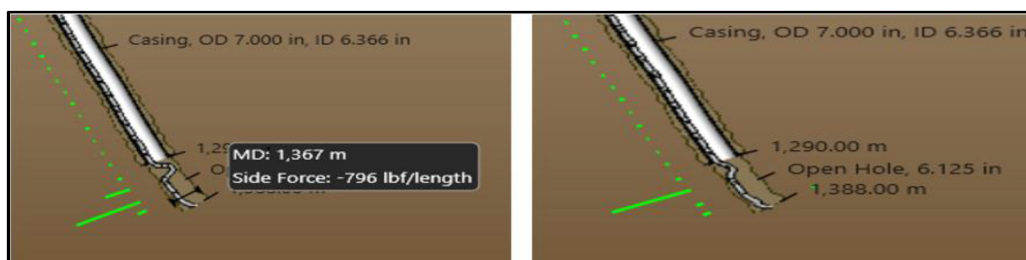


Figure 18—Simulation of 6m and 12m gun using 4 3/4" Drill Collar Joints, demonstrating improved performance with the TCP BHA reaching TD without restrictions.

Step 3 - Second Dummy Run: 6 m (20 ft) Gun Assembly

In line with the simulation findings, four joints of 4 3/4" heavy-weight drill collar (HWDC) were added above the gun assembly (Figure 19). While a second dummy run with a 12 m gun was proposed to further validate conveyance for a longer assembly, the customer elected to proceed with the 6 m configuration to avoid additional operational time. This setup successfully reached the planned target depth of 1,388 m to tag the 6 1/8" hole TD without restriction, confirming the suitability of the revised string design for live perforation runs.

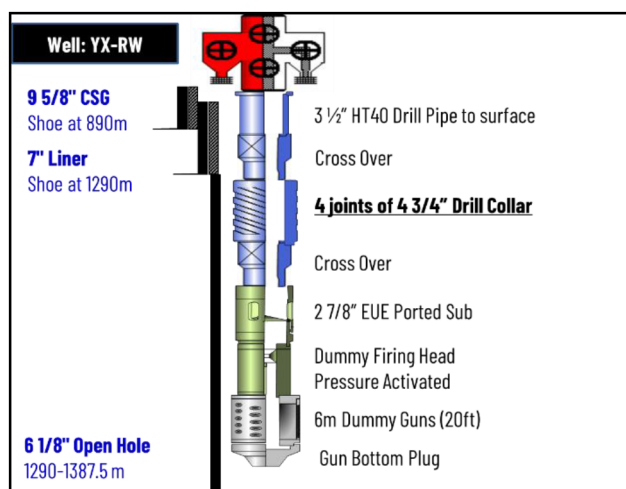


Figure 19—Bottom-hole assembly setup for TCP during second dummy run in Well YX-RW with heaving drill collar added.

Step 4 - Final Perforation Design

Simulation results indicated that running a single 6 m gun in the worst-case phasing scenario could result in zero shots impacting the 7" casing. As the original perforation strategy of deploying three oriented joints was not feasible under the prevailing wellbore conditions, an alternative configuration was selected: two 3 m guns with a 20-30° orientation offset, as shown in Figure 20. This arrangement improved the worst-case prediction to approximately 10 shots impacting the target casing.

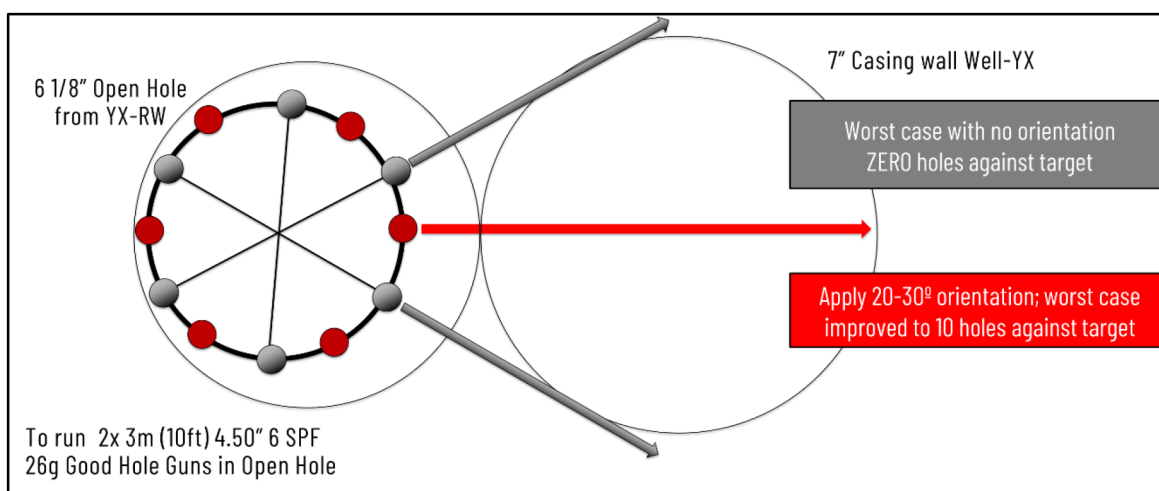


Figure 20—Revised design for worst-case scenario, based on dummy run results, to improve target coverage by applying 20-30° shift between guns.

In the best-case scenario, applying the 20-30° orientation offset could increase the number of impacts to around 20 shots (Figure 21); however, this outcome remains dependent on the final gun landing position prior to firing.

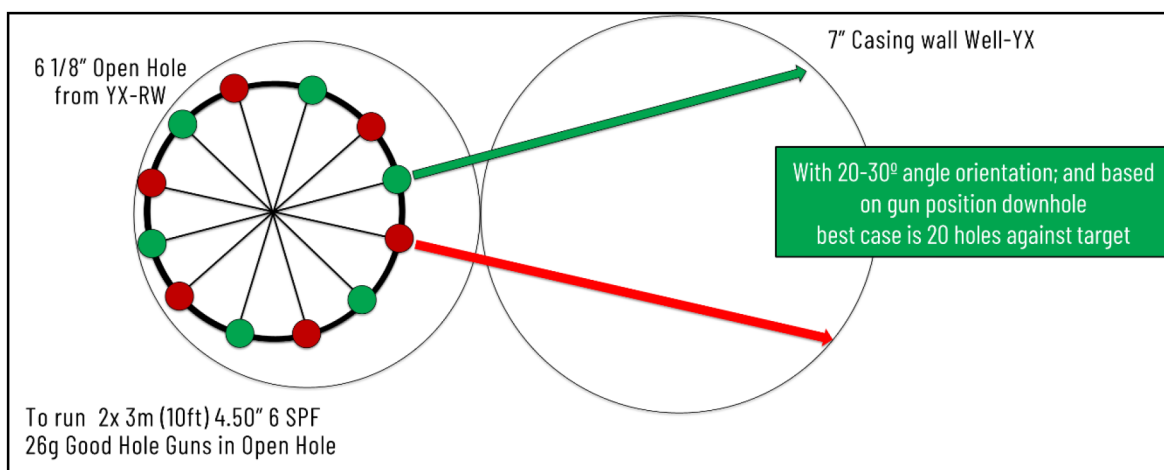


Figure 21—Revised design illustrating the best-case scenario to achieve maximum shots against 7" casing walls.

Step 5 - Perforation BHA Configuration for Execution

The final TCP string configuration incorporated findings from both dummy runs and the simulated tally generated using EDM software. To maximize the probability of successful casing penetration, two 3 m guns were deployed with a 25° orientation offset between them, supported by four joints of 4 3/4" heavy-weight drill collars (HWDC) installed above the gun assembly (Figure 22). The perforation operation was executed twice to cover the full 10 m target interval, with a 2-3 m overlap applied during the second run to ensure complete coverage.

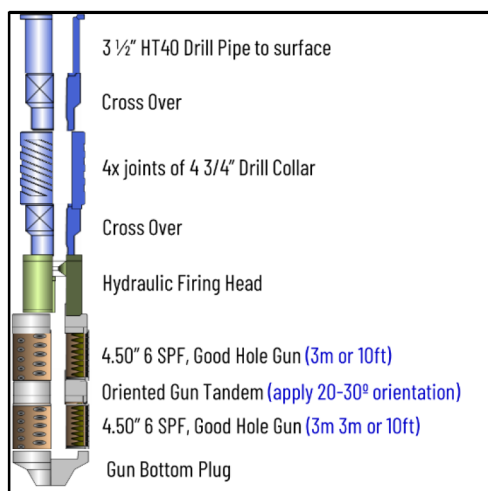


Figure 22—Final Tubing Conveyed Perforation Bottom Hole Assembly

During POOH after the first run, no fluid losses were observed, prompting concern that the second run might also fail to establish communication. To mitigate this risk and avoid potential delays given the one-day transport time from the gun loading facility to the rig, the customer instructed the preparation and dispatch of a third set of guns while the second run was being planned. Following the second run, significant fluid losses were observed, confirming communication with the target well, and the third run was subsequently cancelled.

Step 6 - Cement Placement for Abandonment

After the second perforation, hydraulic communication was successfully established with the target wellbore, enabling cement placement operations. A total of 8-9 m³ of cement was pumped through the

perforated casing, ensuring effective coverage across the interval and restoring annular isolation. As direct well access was not available, conventional CBL or ultrasonic verification could not be performed.

SNL logs were acquired both before and after the perforation and cement placement (Figure 23). The pre-job SNL clearly showed crossflow between the Shuaiba and Natih formations, whereas the post-job SNL confirmed that the crossflow had been eliminated. This direct comparison demonstrated that the seal was successfully achieved and validated the effectiveness of the abandonment operation through side-wall perforation from the relief well, ensuring long-term well integrity.

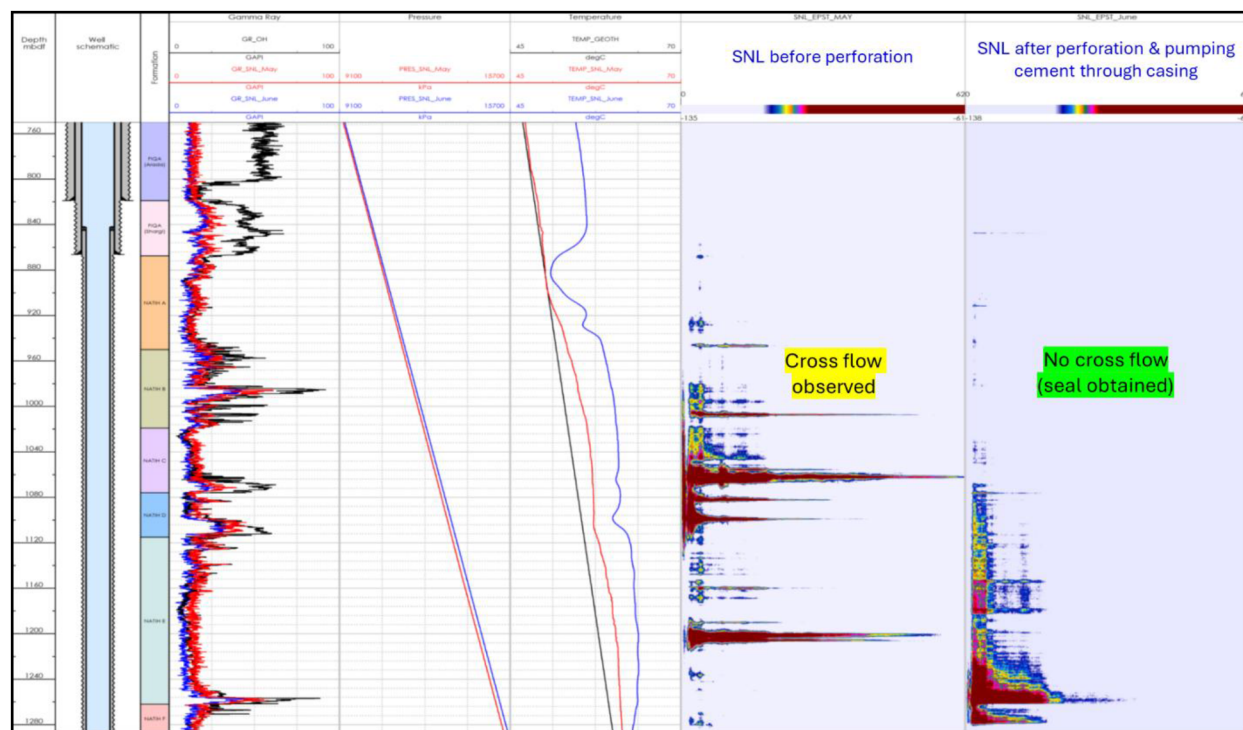


Figure 23—SNL logs before and after perforation and cement placement, showing elimination of crossflow.

CONCLUSION

The perforation program achieved its primary objective, with the 7" casing wall successfully perforated and significant fluid losses observed immediately following the second run. These losses confirmed the establishment of communication between the relief well and the target well. The application of phased orientation between guns, based on simulation and coupon test findings, increased the probability of impacting the target casing wall and improved coverage across the designed interval. This configuration created multiple penetration points for cement placement, ensuring effective slurry entry and enhanced bonding, and ultimately the establishment of a permanent seal across the abandonment interval.

Lessons learned include the critical role of coupon testing in validating gun performance under realistic conditions, the effectiveness of phased orientation in increasing punching casing wall success probability, and the need for flexible planning when initial interception methods fail.

In terms of applicability, this methodology provides a viable option for future high-risk abandonment scenarios where direct access to the target well is impossible, particularly in mature fields or complex wellbores where conventional interventions are not feasible.

The main limitations of the approach are its reliance on available gun sizes and orientation tools, challenges in wells with extreme dogleg severity or tight restrictions, and the operational complexity of executing side-wall perforation from an open hole.

This operation represents the first documented case in the history of perforation where external side-wall perforation was successfully executed through a relief well to abandon an inaccessible target well, achieving long-term annular isolation and providing a proven reference for similar high-risk abandonment projects worldwide.

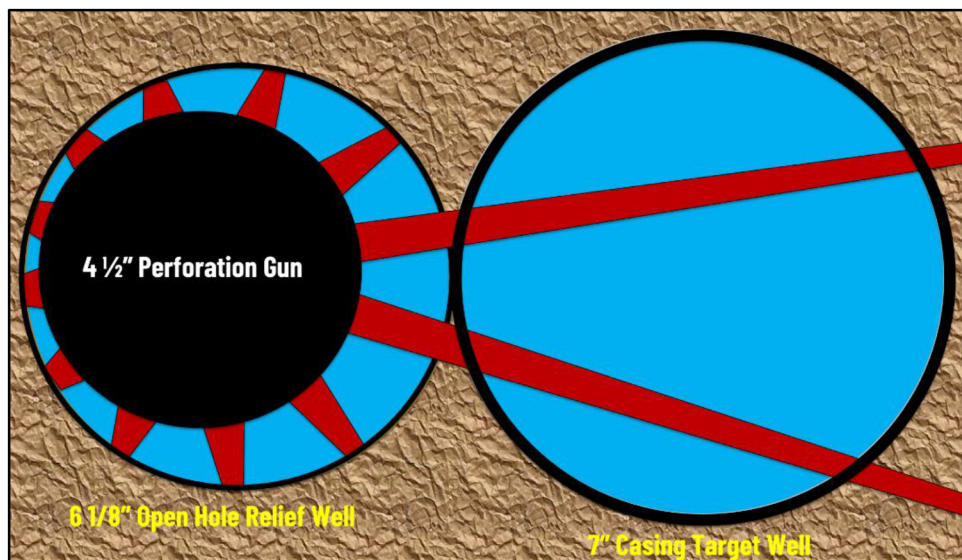


Figure 24—Schematic representation of the first documented side-wall perforation from an open-hole relief well, successfully penetrating two walls of the 7" target casing to establish communication for abandonment.

BEST PRACTICES AND RECOMMENDATIONS

The unique challenges encountered during the Well-YX relief well perforation program ranging from degraded tubular integrity and limited well access to high annular pressures and uncertainty in tool positioning required an integrated approach that combined sound engineering design with adaptive field execution. The solutions developed under these extreme conditions provide a valuable reference for future high-risk abandonment projects where conventional methods are not feasible.

The following best practices are drawn from the lessons learned in this operation and are presented to assist in planning and executing similar squeeze operations. They address both well design and interception considerations, which influence the overall feasibility of establishing communication with the target well, and perforation configuration guidelines, which directly affect the success of cement placement and zonal isolation

1. Well Design & Interception

- Drill the relief well parallel to the target well with minimal clearance to improve interception accuracy.
- Maintain a predominantly vertical profile with low dogleg severity to enhance tool conveyance, reduce mechanical risks, and improve control over the perforation process.
- Where feasible, plan the interception in a cased-hole section preferably with a single casing string to improve operational stability, minimize hole instability, and allow more precise control of perforation depth and phasing.

2. Perforation Configuration

- For optimal performance under favorable wellbore conditions, deploy a 5.00" oriented gun with 8 shots per foot (SPF), loaded with 26 g Good Hole charges, and utilizing line-shaped phasing to maximize casing penetration.

- Apply a controlled orientation between guns to increase the number of shots impacting the target casing wall, improving cement entry points and ensuring more uniform coverage.
- Use accurate depth correlation and careful alignment of phasing to maximize the probability of achieving full isolation in a single perforation run.

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